

# Constrained Load Transportation by A Team of Quadrotors

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**Abstract**—Cable-suspended load transportation by multiple unmanned aerial vehicles (UAVs) has many applications. However, system nonlinearities and uncertainties as well as load dimensions are usually ignored by the literature, while the performance and safety constraints during the cooperative transportation operation are not discussed by existing works. In this work, we develop a new adaptive constrained formation control architecture for a team of UAVs that are collaboratively transporting a load, subject to multiple performance and safety constraint requirements. Universal barrier functions are used to address these time-varying constraints, and adaptive estimators are employed to deal with system uncertainties. In the end, a simulation study further demonstrates the effectiveness of the proposed framework.

## I. INTRODUCTION

Cable-suspended load transportation by unmanned aerial vehicles (UAVs), especially quadrotors, has attracted significant attention over recent years. However, a single UAV usually suffers from limited payload capacity, and is prone to failure in face of mechanical breakdowns. Therefore, it is advantageous to use a group of UAVs to collaboratively transport a common payload together.

There are several challenges associated with the cooperative load transportation task. First is the nonlinearities and uncertainties in the UAV model and the dimension of the load. Most existing works assume a linear model of UAVs, as in [1], [2], or assume the load is simply a point mass, as in [3]–[7]. Such frameworks do not take the highly nonlinear UAV model into considerations, and are not suitable for a wide range of practical applications where the load may have irregular dimensions.

Another challenge is associated with multiple operational constraints on performance and safety during the cooperative load transport. On one hand, the UAV team and load need to stay close to the desired path, so that to ensure desirable formation pattern and avoid collision with nearby obstacles. On the other hand, the distance between any two UAVs in the team cannot be either too small or too large, which can lead to inter-UAV collision and breakdown of the suspension cable, respectively. Failure to satisfy these performance and safety constraint requirements will lead to mission failures and/or loss of UAVs and load. Existing cooperative load transport algorithms, such as those presented in [8]–[11], cannot take such performance and safety constraint requirements into consideration. Conventional safety-critical UAV formation algorithms, such as those discussed in [12]–[17], are not applicable in the load transportation task, due to the presence of cable-suspended load and multiple yet different constraint requirements.

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In this work, we propose and investigate a new adaptive constrained formation control architecture for a team of UAVs that are collaboratively transporting a cable-suspended rigid-body load. Specifically, to ensure operation performance, we consider constraint requirements on distance tracking errors of the payload and each UAV. To guarantee safety, we address safety constraints on the inter-UAV distances, which cannot be either too large or too small. Universal barrier functions first proposed in our earlier works [18]–[20] are used to deal with the constraint requirements, which can address different types of constraints in a unified framework. Through rigorous analysis, we show that exponential convergence can be guaranteed on the distance tracking errors, with all constraint requirements satisfied during the load transportation operation.

## II. PROBLEM FORMULATION

### A. System Dynamics

Consider a group of  $N$  ( $N \geq 3$ ) UAVs that are cooperatively transporting a rigid body payload connected via cables, where the dynamics of the UAVs and the load are given as

$$\text{UAVs} \begin{cases} m_i \ddot{p}_i(t) = -F_i(t)R(\Theta_i(t))e_z + m_i g e_z \\ \quad + T_i(t)R(\Theta_L(t))e_i(t), \end{cases} \quad (1)$$

$$\text{Load} \begin{cases} m_L \ddot{p}_L(t) = m_L g e_z - \sum_{i=1}^N T_i(t)R(\Theta_L(t))e_i(t) \\ \dot{\Theta}_L(t) = \Gamma(\Theta_L(t))\omega_L(t) \\ J_L \dot{\omega}_L(t) + \mathbb{S}(\omega_L(t))J_L \omega_L(t) \\ \quad = \sum_{i=1}^N \mathbb{S}(r_i)(-T_i(t)e_i(t)), \end{cases} \quad (2)$$

where  $m_i \in \mathbb{R}$ ,  $m_i > 0$  is the mass of the  $i$ th quadrotor ( $i = 1, \dots, N$ ). The position and attitude in the inertial reference frame are represented as  $p_i(t) = [x_i(t), y_i(t), z_i(t)]^T \in \mathbb{R}^3$  and  $\Theta_i(t) = [\phi_i(t), \theta_i(t), \psi_i(t)]^T \in \mathbb{R}^3$ , respectively.  $R(\Theta_i(t)) \in \text{SO}(3)$  is the rotation matrix, which relates the body-fixed frame to the inertial frame and is expressed as

$$R(\Theta_i) = \begin{bmatrix} c\theta_i c\psi_i & s\phi_i s\theta_i c\psi_i - c\phi_i s\psi_i & c\phi_i s\theta_i s\psi_i + s\phi_i s\psi_i \\ c\theta_i s\psi_i & s\phi_i s\theta_i s\psi_i + c\phi_i c\psi_i & c\phi_i s\theta_i c\psi_i - s\phi_i c\psi_i \\ -s\theta_i & s\phi_i c\theta_i & c\phi_i c\theta_i \end{bmatrix}. \quad (3)$$

Furthermore,  $g \in \mathbb{R}$  is the gravitational acceleration and  $e_z = [0, 0, 1]^T \in \mathbb{R}^3$  is the unit vector. Finally,  $T_i(t) \in \mathbb{R}$  represents the tension in the  $i$ th rigid cable,  $F_i(t) \in \mathbb{R}$  represents the thrust of the  $i$ th quadrotor ( $i = 1, \dots, N$ ), respectively.

Next,  $m_L \in \mathbb{R}^+$  and  $J_L \in \mathbb{R}^{3 \times 3}$  are the load mass and inertia that is symmetric and positive definite, where the subscript L stands for “Load”.  $p_L(t) = [x_L(t), y_L(t), z_L(t)]^T \in$

$\mathbb{R}^3$  and  $\Theta_L(t) = [\phi_L(t), \theta_L(t), \psi_L(t)]^T \in \mathbb{R}^3$  represent the load position and attitude in the inertial reference frame, and  $\omega_L(t) = [\omega_{xL}(t), \omega_{yL}(t), \omega_{zL}(t)]^T \in \mathbb{R}^3$  represents the load rotational velocity with respect to its body-fixed frame. Furthermore,  $r_i \in \mathbb{R}^3$  is the attachment point on the payload by the  $i$ th link, represented in the payload body-fixed frame. Moreover,  $e_i(t) \in \mathbb{S}^2$  is the unit direction vector from the mass center of the  $i$ th UAV towards the  $i$ th link attachment point.  $\Gamma(\Theta_L(t))$  is the transformation matrix that relates the angular velocity in the body-fixed frame to the rate of change of the Euler angles in the inertial frame, and is given by

$$\Gamma(\Theta_L) = \begin{bmatrix} 1 & s\phi_L t\theta_L & c\phi_L t\theta_L \\ 0 & c\phi_L & -s\phi_L \\ 0 & s\phi_L/c\theta_L & c\phi_L/c\theta_L \end{bmatrix}, \quad (4)$$

which is well defined and invertable when  $-\frac{\pi}{2} < \phi_L(t) < \frac{\pi}{2}$  and  $-\frac{\pi}{2} < \theta_L(t) < \frac{\pi}{2}$ .

Denote  $\Psi(\Theta_L(t)) = \Gamma^{-1}(\Theta_L(t))$ , from (1) we get  $\omega_L(t) = \Psi(\Theta_L(t))\dot{\Theta}_L(t)$ . Hence, multiply  $J_L^T$  on both sides of the third equation in (2), we can obtain

$$M_L(\Theta_L(t))\ddot{\Theta}_L(t) + C_L(\Theta_L(t), \dot{\Theta}_L(t))\dot{\Theta}_L(t) = \Psi^T(\Theta_L(t))J_L^T \sum_{i=1}^N \mathbb{S}(r_i)(-T_i(t)e_i(t)), \quad (5)$$

where  $M_L(\Theta_L(t)) = \Psi^T(\Theta_L(t))J_L^T J_L \Psi(\Theta_L(t))$ , and  $C_L(\Theta_L(t), \dot{\Theta}_L(t)) = \Psi^T(\Theta_L(t))J_L^T \mathbb{S}(\Psi(\Theta_L(t))\dot{\Theta}_L(t))J_L \Psi(\Theta_L(t)) + \Psi^T(\Theta_L(t))J_L^T J_L \dot{\Psi}(\Theta_L(t))$ .

### B. System Performance and Safety Constraints

In the cooperative transportation task, the payload is supposed to track its desired trajectory, denoted by  $p_{dL}(t) \triangleq [x_{dL}(t), y_{dL}(t), z_{dL}(t)]^T \in \mathbb{R}^3$ . Moreover, all UAVs need to track a desired formation pattern, with the coordinate of the reference trajectory for the  $i$ th vehicle ( $i = 1, \dots, N$ ) denoted by  $p_{di}(t) \triangleq [x_{di}(t), y_{di}(t), z_{di}(t)]^T \in \mathbb{R}^3$ .

Now, define the line-of-sight (LOS) distance tracking error for the payload  $d_{eL}(t)$  as

$$d_{eL} \triangleq \sqrt{(x_L - x_{dL})^2 + (y_L - y_{dL})^2 + (z_L - z_{dL})^2}, \quad (6)$$

which is the distance between the desired and actual position of the payload. Furthermore, for the  $i$ th quadrotor ( $i = 1, \dots, N$ ), define the line-of-sight distance tracking error  $d_{ei}(t)$  as

$$d_{ei} \triangleq \sqrt{(x_i - x_{di})^2 + (y_i - y_{di})^2 + (z_i - z_{di})^2}. \quad (7)$$

Besides, the desired LOS relative distance between any two quadrotors  $\vartheta_{ij}(t)$  is given as

$$\vartheta_{ij} \triangleq \sqrt{(x_{di} - x_{dj})^2 + (y_{di} - y_{dj})^2 + (z_{di} - z_{dj})^2}, \quad (8)$$

and the actual LOS relative distance  $d_{ij}(t)$  is

$$d_{ij} \triangleq \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2}. \quad (9)$$

During the cooperative transportation, there are certain **system constraint requirements** that need to be satisfied, in order to ensure the *precise* and *safe* functioning of the system. First, the payload distance tracking error  $d_{eL}(t)$  has to meet the *performance constraint* requirement that

$$d_{eL}(t) < \Omega_{dHL}(t), \quad (10)$$

where, for all  $t \geq 0$ ,  $\Omega_{dHL}(t) > 0$  is the user-defined time-varying constraint requirement function and is  $C^3$ . This means the payload should not deviate much from its desired trajectory. Moreover, the LOS distance tracking error for the  $i$ th quadrotor  $d_{ei}(t)$  ( $i = 1, \dots, N$ ) has to satisfy the following *performance constraint*

$$d_{ei}(t) < \Omega_{dHi}(t), \quad (11)$$

where, for all  $t \geq 0$ ,  $\Omega_{dHi}(t) > 0$  is the user-defined time-varying constraint requirement function and is  $C^3$ .

Next, define the LOS relative distance tracking error between the  $i$ th and  $j$ th quadrotors ( $i, j = 1, \dots, N, j \neq i$ ) as  $d_{eij}(t) \triangleq d_{ij}(t) - \vartheta_{ij}(t)$ , which has to meet the following *safety constraint*

$$-\Omega_{Wij}(t) < d_{eij}(t) < \Omega_{Hij}(t), \quad (12)$$

where, for all  $t \geq 0$ ,  $\Omega_{Hij}(t) > 0$  is the user-defined time-varying upper bound for the distance tracking error  $d_{eij}(t)$ , and  $-\Omega_{Wij}(t) < 0$  is the lower bound, with  $\vartheta_{ij}(t) > \Omega_{Wij}(t) > 0$ . Both  $\Omega_{Hij}(t)$  and  $\Omega_{Wij}(t)$  are  $C^3$ .

### C. Control Objective

The **control objective** for the cooperative transportation problem is to design a control framework such that:

- 1) The payload distance tracking error  $d_{eL}(t)$  and the UAV distance tracking errors  $d_{ei}(t)$  ( $i = 1, \dots, N$ ) can all converge into arbitrarily small neighbourhoods of zero;
- 2) The relative distance error  $d_{eij}(t)$  between the  $i$ th and  $j$ th ( $i, j = 1, \dots, N, j \neq i$ ) quadrotors can converge into arbitrarily small neighbourhoods of zero;
- 3) The payload attitude tracking error  $\Theta_L(t) - \Theta_{dL}(t)$  can converge into arbitrarily small neighbourhoods of zero;
- 4) The performance and safety constraint requirements (10), (11), and (12) are satisfied during the operation.

The following assumptions are used to facilitate the discussion and analysis of the main result.

**Assumption 2.1:** The reference trajectory coordinates for the payload  $x_{dL}(t)$ ,  $y_{dL}(t)$ , and  $z_{dL}(t)$  are all  $C^3$ , and for the  $i$ th quadrotor ( $i = 1, \dots, N$ )  $x_{di}(t)$ ,  $y_{di}(t)$ , and  $z_{di}(t)$  are  $C^3$ . Moreover, the reference payload attitude  $\Theta_{dL}(t)$  is  $C^3$ .

**Assumption 2.2:** The thrust  $F_i(t)$  for the  $i$ th quadrotor ( $i = 1, \dots, N$ ) is uniformly bounded with *unknown* bounds. Moreover, the cables are massless and cannot be stretched, and the tensions in the cables are *unknown* but bounded.

**Assumption 2.3:** The payload mass  $m_L$  and UAV mass  $m_i$  for the  $i$ th quadrotor ( $i = 1, \dots, N$ ) are known. However, the payload inertia  $J_L$  is *unknown*, and is assumed to be both upper and lower bounded, such that for any  $z \in \mathbb{R}^3$ ,  $\bar{J}_L z^T z > z^T J_L z > \underline{J}_L z^T z$  where  $\bar{J}_L$  and  $\underline{J}_L$  are *unknown* positive constants.

**Assumption 2.4:** Let

$$P = \begin{bmatrix} I_3 & I_3 & \cdots & I_3 \\ \mathbb{S}(r_1) & \mathbb{S}(r_2) & \cdots & \mathbb{S}(r_N) \end{bmatrix} \in \mathbb{R}^{6 \times 3N}, \quad (13)$$

we assume  $\text{rank}(P) = 6$ .

**Remark 2.1:** This full row rank assumption in Assumption 2.4 can be realized when  $N \geq 3$ .

**Assumption 2.5:** The UAV and load attitudes satisfy  $-\frac{\pi}{2} < \phi_L(t) < \frac{\pi}{2}$ ,  $-\frac{\pi}{2} < \theta_L(t) < \frac{\pi}{2}$ ,  $-\frac{\pi}{2} < \phi_i(t) < \frac{\pi}{2}$ , and  $-\frac{\pi}{2} < \theta_i(t) < \frac{\pi}{2}$ , where  $i = 1, \dots, N$ .

To facilitate the analysis, we present the following lemma.

**Lemma 2.1:** For any constant  $\varepsilon > 0$  and any variable  $z \in \mathbb{R}$ , we have  $0 \leq |z| - \frac{z^2}{\sqrt{z^2 + \varepsilon^2}} < \varepsilon$ .

In order to simplify representations of signals, we will omit the time and state dependence of signals for the rest of the discussion in this work.

### III. UNIVERSAL BARRIER FUNCTION

To address the constraint requirements (10), (11), and (12), we first introduce transformed error variables for the load and the  $i$ th quadrotor ( $i = 1, \dots, N$ ) as follows

$$\eta_{eL} = \frac{\Omega_{dHL} d_{eL}}{\Omega_{dHL} - d_{eL}}, \quad \eta_{ei} = \frac{\Omega_{dHi} d_{ei}}{\Omega_{dHi} - d_{ei}}, \quad (14)$$

and

$$\eta_{eij} = \frac{\Omega_{Hi} \Omega_{Wij} d_{eij}}{(\Omega_{Hi} - d_{eij})(\Omega_{Wij} + d_{eij})}. \quad (15)$$

The universal barrier functions used to deal with the constraint requirements (10), (11), and (12) for the  $i$ th quadrotor ( $i = 1, \dots, N$ ) are then defined as

$$V_{eL} = \frac{1}{2} \eta_{eL}^2, \quad V_{ei} = \frac{1}{2} \eta_{ei}^2, \quad V_{eij} = \frac{1}{2} \eta_{eij}^2. \quad (16)$$

### IV. CONTROL DESIGN AND ANALYSIS

#### A. “Desired Cable Tension” Design

##### Step 1:

We first consider the position kinematics of the payload. The time derivative of the UBF  $V_{eL}$  leads to

$$\dot{V}_{eL} = \eta_{eL} \left[ \frac{\partial \eta_{eL}}{\partial \Omega_{dHL}} \dot{\Omega}_{dHL} + \frac{\partial \eta_{eL}}{\partial d_{eL}} \frac{1}{d_{eL}} \left( (x_L - x_{dL}) \dot{x}_L + (y_L - y_{dL}) \dot{y}_L + (z_L - z_{dL}) \dot{z}_L - \xi_L \right) \right], \quad (17)$$

where  $\xi_L = (x_L - x_{dL}) \dot{x}_{dL} + (y_L - y_{dL}) \dot{y}_{dL} + (z_L - z_{dL}) \dot{z}_{dL}$ . Now, denote  $E_L = \frac{1}{d_{eL}} [x_L - x_{dL}, y_L - y_{dL}, z_L - z_{dL}]^T \in \mathbb{R}^3$ . Let  $z_{2L} = \dot{p}_L - \alpha_{pL}$ , we design the stabilizing function as

$$\alpha_{pL} = \frac{E_L}{\frac{\partial \eta_{eL}}{\partial d_{eL}}} \left( -\frac{\partial \eta_{eL}}{\partial \Omega_{dHL}} \dot{\Omega}_{dHL} - K_{1L} \eta_{eL} \right) + \dot{p}_{dL}, \quad (18)$$

where  $\frac{\partial \eta_{eL}}{\partial d_{eL}} = \frac{\Omega_{dHL}^2}{(\Omega_{dHL} - d_{eL})^2} > 0$  is always guaranteed, and  $K_{1L} > 0$  is a control constant. (17) further leads to

$$\dot{V}_{eL} = \eta_{eL} \frac{\partial \eta_{eL}}{\partial d_{eL}} E_L^T z_{2L} - K_{1L} \eta_{eL}^2. \quad (19)$$

##### Step 2:

Now we address the translational dynamics of the payload. Design the Lyapunov function candidate as  $V_{2L} = \frac{1}{2} m_L z_{2L}^T z_{2L}$ , and its time derivative leads to

$$\dot{V}_{2L} = z_{2L}^T \left[ -\sum_{i=1}^N T_{di} R_{dL} e_{di} - \sum_{i=1}^N T_{di} (R_L - R_{dL}) e_{di} + \sum_{i=1}^N R_L (T_{di} e_{di} - T_i e_i) + m_L g e_z - m_L \dot{\alpha}_{pL} \right]. \quad (20)$$

Under Assumption 2.3 and Lemma 2.1, we have  $z_{2L}^T \left[ -\sum_{i=1}^N T_{di} (R_L - R_{dL}) e_{di} + \sum_{i=1}^N R_L (T_{di} e_{di} - T_i e_i) \right] \leq \|z_{2L}\| \bar{\delta}_{1L} < \varepsilon_L \bar{\delta}_{1L} + \bar{\delta}_{1L} \frac{\|z_{2L}\|^2}{\sqrt{\|z_{2L}\|^2 + \varepsilon_L^2}}$ , with  $\bar{\delta}_{1L} > 0$  being an unknown upper bound, and  $\varepsilon_L$  being a small positive constant. Next, treat  $-\sum_{i=1}^N T_{di} R_{dL} e_{di}$  in (20) as a virtual control signal and design it as

$$\begin{aligned} -\sum_{i=1}^N T_{di} R_{dL} e_{di} &= -F_{dL} \\ &= -m_L g e_z + m_L \dot{\alpha}_{pL} - K_{2L} z_{2L} - \eta_{eL} \frac{\partial \eta_{eL}}{\partial d_{eL}} E_L \\ &\quad - \hat{\delta}_{1L} \frac{z_{2L}}{\sqrt{\|z_{2L}\|^2 + \varepsilon_L^2}}, \end{aligned} \quad (21)$$

where  $K_{2L} > 0$  is a control constant, and  $\hat{\delta}_{1L}$  is the adaptive estimate of the unknown constant  $\bar{\delta}_{1L}$ , which is designed as

$$\dot{\hat{\delta}}_{1L} = n_{\delta 1L} \frac{\|z_{2L}\|^2}{\sqrt{\|z_{2L}\|^2 + \varepsilon_L^2}} - \sigma_{\delta 1L} \hat{\delta}_{1L}, \quad (22)$$

with  $n_{\delta 1L}$  and  $\sigma_{\delta 1L}$  being positive adaptive gains.

Now, design the Lyapunov function as  $V_{L_{oad1}} = V_{eL} + V_{2L} + V_{\delta 1L}$ ,  $V_{\delta 1L} = \frac{1}{2n_{\delta 1L}} \bar{\delta}_{1L}^2$ , where  $\bar{\delta}_{1L} = \hat{\delta}_{1L} - \bar{\delta}_{1L}$ , and from (19), (20), (21), and (22), we can get

$$\dot{V}_{Load1} < -K_{1L} \eta_{eL}^2 - K_{2L} z_{2L}^T z_{2L} - \frac{\sigma_{\delta 1L}}{2n_{\delta 1L}} \bar{\delta}_{1L}^2 + c_{1L}, \quad (23)$$

where  $c_{1L} \triangleq \left( \varepsilon_L \bar{\delta}_{1L} + \frac{\sigma_{\delta 1L}}{2n_{\delta 1L}} \bar{\delta}_{1L}^2 \right)$  is a constant.

##### Step 3:

Next, we discuss the attitude kinematics of the payload. Define  $z_{3L} = \Theta_L - \Theta_{dL}$  and  $z_{4L} = \dot{\Theta}_L - \alpha_{\Theta L}$ , where the stabilizing function is designed as

$$\alpha_{\Theta L} = -K_{3L} z_{3L} + \dot{\Theta}_{dL}, \quad (24)$$

where  $K_{3L} > 0$  is a control constant. Now, design the Lyapunov function candidate as  $V_{3L} = \frac{1}{2} z_{3L}^T z_{3L}$ , its time derivative leads to

$$\dot{V}_{3L} = -K_{3L} z_{3L}^T z_{3L} + z_{3L}^T z_{4L}. \quad (25)$$

##### Step 4:

Finally, for the load dynamics control, we design the Lyapunov function candidate as  $V_{4L} = \frac{1}{2} z_{4L}^T M_L z_{4L}$ . With some algebraic analysis shown in Appendix A, treat  $\sum_{i=1}^N \mathbb{S}(r_i) (-T_{di} e_{di})$  in (48) as a virtual control signal, represent it as  $-M_{dL}$  for convenience, and design it as

$$\begin{aligned} \sum_{i=1}^N \mathbb{S}(r_i) (-T_{di} e_{di}) &= -M_{dL} \\ &= -\frac{\Psi_L z_{4L} \|\bar{\mu}_L\|^2 \hat{\rho}_{JL}^2}{\sqrt{\|\Psi_L z_{4L}\|^2 \|\bar{\mu}_L\|^2 \hat{\rho}_{JL}^2 + \varepsilon_L^2}}, \end{aligned} \quad (26)$$

where

$$\bar{\mu}_L = K_{4L} \Gamma_L^T z_{4L} + \Gamma_L^T z_{3L} + \hat{\zeta}_L \frac{\Gamma_L^T z_{4L} \Xi_L^2}{\sqrt{\|z_{4L}\|^2 \Xi_L^2 + \varepsilon_L^2}}$$

$$+ \hat{\delta}_{2L} \frac{\Gamma_L^T z_{4L}}{\sqrt{\|z_{4L}\|^2 + \varepsilon_L^2}}. \quad (27)$$

Here,  $K_{4L} > 0$  is a positive control gain.  $\Xi_L$  is introduced in (50).  $\hat{\rho}_{JL}$  is the adaptive estimate of the unknown lower bound  $\rho_{JL} = \frac{1}{J_L}$ ,  $\hat{\zeta}_L$  is the adaptive estimate of the unknown bound  $\bar{\zeta}_L$  that is introduced in (50), and  $\hat{\delta}_{2L}$  is the adaptive estimate of the unknown bound  $\bar{\delta}_{2L}$  that is introduced in (49), where the adaptive laws are designed as

$$\dot{\hat{\rho}}_{JL} = n_{\rho JL} z_{4L}^T \Psi_L^T \bar{\mu}_L - \sigma_{\rho JL} \hat{\rho}_{JL}, \quad (28)$$

$$\dot{\hat{\zeta}}_L = n_{\zeta L} \frac{\|z_{4L}\|^2 \Xi_L^2}{\sqrt{\|z_{4L}\|^2 \Xi_L^2 + \varepsilon_L^2}} - \sigma_{\zeta L} \hat{\zeta}_L, \quad (29)$$

$$\dot{\hat{\delta}}_{2L} = n_{\delta 2L} \frac{\|z_{4L}\|^2}{\sqrt{\|z_{4L}\|^2 + \varepsilon_L^2}} - \sigma_{\delta 2L} \hat{\delta}_{2L}, \quad (30)$$

with  $n_{\rho JL}$ ,  $\sigma_{\rho JL}$ ,  $n_{\zeta L}$ ,  $\sigma_{\zeta L}$ ,  $n_{\delta 2L}$ , and  $\sigma_{\delta 2L}$  are positive adaptive gains. Next, we design the Lyapunov function candidate as  $V_{\text{Load2}} = V_{3L} + V_{4L} + V_{\zeta L} + V_{\rho L} + V_{\delta 2L}$ ,  $V_{\zeta L} = \frac{1}{2n_{\zeta L}} \tilde{\zeta}_L^2$ ,  $V_{\rho L} = \frac{J_L}{2n_{\rho JL}} \tilde{\rho}_{JL}^2$ ,  $V_{\delta 2L} = \frac{1}{2n_{\delta 2L}} \tilde{\delta}_{2L}^2$ , where  $\tilde{\zeta}_L = \hat{\zeta}_L - \bar{\zeta}_L$ ,  $\tilde{\rho}_{JL} = \hat{\rho}_{JL} - \rho_{JL}$  ( $\rho_{JL} = \frac{1}{J_L}$ ), and  $\tilde{\delta}_{2L} = \hat{\delta}_{2L} - \bar{\delta}_{2L}$ . The time derivative of  $V_{\text{Load2}}$  leads to

$$\begin{aligned} \dot{V}_{\text{Load2}} &< -K_{3L} z_{3L}^T z_{3L} - K_{4L} z_{4L}^T z_{4L} - \frac{\sigma_{\zeta L}}{2n_{\zeta L}} \tilde{\zeta}_L^2 \\ &\quad - \frac{\sigma_{\rho JL} J_L}{2n_{\rho JL}} \tilde{\rho}_{JL}^2 - \frac{\sigma_{\delta 2L}}{2n_{\delta 2L}} \tilde{\delta}_{2L}^2 + c_{2L}, \end{aligned} \quad (31)$$

where  $c_{2L}$  is a constant and is defined as  $c_{2L} \triangleq \varepsilon_L J_L + \varepsilon_L (\bar{\delta}_{2L} + \bar{\zeta}_L) + \frac{\sigma_{\zeta L}}{2n_{\zeta L}} \bar{\zeta}_L^2 + \frac{\sigma_{\rho JL}}{2n_{\rho JL}} \frac{1}{J_L} + \frac{\sigma_{\delta 2L}}{2n_{\delta 2L}} \bar{\delta}_{2L}^2$ .

#### Step 5:

Now the “desired cable tension” is derived from (21) and (26)

$$\begin{bmatrix} T_{d1} e_{d1} \\ \vdots \\ T_{dN} e_{dN} \end{bmatrix} = P^T (P P^T)^{-1} \begin{bmatrix} R_{dL}^T F_{dL} \\ M_{dL} \end{bmatrix}. \quad (32)$$

### B. UAV Distance Control Design

#### Step 6:

Define  $V_1 = \sum_{i=1}^N (V_{ei} + \sum_{j=1, j \neq i}^N V_{eij})$ , with  $V_{ei}$  and  $V_{eij}$  defined in (16). With algebraic analysis shown in Appendix B, for  $\dot{V}_1$  we have

$$\dot{V}_1 = \sum_{i=1}^N \left\{ -\eta_{ei} \xi_i - \sum_{j=1, j \neq i}^N \eta_{eij} \xi_{eij} + E_i^T \dot{p}_i \right\}, \quad (33)$$

where  $\xi_i$  is introduced in (51),  $\xi_{eij}$  is introduced in (52), and  $E_i = [E_{xi}, E_{yi}, E_{zi}]^T \in \mathbb{R}^3$  with  $E_{\varpi i} = \eta_{ei} \frac{\partial \eta_{ei}}{\partial d_{ei}} \frac{1}{d_{ei}} (\varpi_i - \varpi_{di}) + \sum_{j=1, j \neq i}^N 2\eta_{eij} \frac{\partial \eta_{eij}}{\partial d_{eij}} \frac{1}{d_{ij}} (\varpi_i - \varpi_j)$ , with  $\varpi = x, y, z$ . Next, define the fictitious velocity tracking error as  $z_{2i} = \dot{p}_i - \alpha_{pi}$ , with the stabilizing function  $\alpha_{pi} \in \mathbb{R}^3$  ( $i = 1, \dots, N$ ) designed as

$$\alpha_{pi} = \frac{E_i}{E_i^T E_i} \left\{ -K_{ei} \eta_{ei}^2 - \sum_{j=1, j \neq i}^N K_{eij} \eta_{eij}^2 \right\}$$

$$+ \eta_{ei} \xi_i + \sum_{j=1, j \neq i}^N \eta_{eij} \xi_{eij} \} \quad (34)$$

where  $K_{ei} > 0$  and  $K_{eij} > 0$  are control gains. Hence, (33) becomes

$$\dot{V}_1 = \sum_{i=1}^N \left( E_i^T z_{2i} - K_{ei} \eta_{ei}^2 - \sum_{j=1, j \neq i}^N K_{eij} \eta_{eij}^2 \right). \quad (35)$$

#### Step 7:

Let  $V_2 = \sum_{i=1}^N \frac{1}{2} m_i z_{2i}^T z_{2i}$ , and its time derivative gives

$$\begin{aligned} \dot{V}_2 &= \sum_{i=1}^N z_{2i}^T \left( -m_i \dot{\alpha}_{pi} + m_i g e_z - u_i + R_L (T_{di} e_{di}^T e_i) e_i \right. \\ &\quad \left. + R_L (T_i - (T_{di} e_{di}^T e_i)) e_i \right), \end{aligned} \quad (36)$$

where we denote  $u_i = F_i R_i e_z$ , with  $R_i \triangleq R(\Theta_i)$ . Now, for the  $i$ th quadrotor ( $i = 1, \dots, N$ ), the control law  $u_i \in \mathbb{R}^3$  is designed as

$$\begin{aligned} u_i &= E_i - m_i \dot{\alpha}_{pi} + m_i g e_z + K_{2i} z_{2i} + \hat{\delta}_i \frac{z_{2i}}{\sqrt{\|z_{2i}\|^2 + \varepsilon_i^2}} \\ &\quad + R_L (T_{di} e_{di}^T e_i) e_i, \end{aligned} \quad (37)$$

$$\dot{\hat{\delta}}_i = n_{\delta i} \frac{\|z_{2i}\|^2}{\sqrt{\|z_{2i}\|^2 + \varepsilon_i^2}} - \sigma_{\delta i} \hat{\delta}_i, \quad (38)$$

where  $K_{2i} > 0$  is a control gain,  $\varepsilon_i > 0$  is a design constant, and  $\hat{\delta}_i$  is the estimation of the unknown constant  $\bar{\delta}_i$  such that under Assumptions 2.2 and 2.3,  $\|R_L (T_i - (T_{di} e_{di}^T e_i)) e_i\| \leq \bar{\delta}_i$ , and  $n_{\delta i}$  and  $\sigma_{\delta i}$  are positive adaptive gains.

Therefore, design the Lyapunov function candidate at this step as  $V_{\text{pos}} = V_1 + V_2 + V_\delta$ ,  $V_\delta = \sum_{i=1}^N \frac{1}{2n_{\delta i}} \tilde{\delta}_i^2$ , where  $\tilde{\delta}_i = \hat{\delta}_i - \bar{\delta}_i$ , and we can further get

$$\begin{aligned} \dot{V}_{\text{pos}} &< \sum_{i=1}^N \left( -K_{ei} \eta_{ei}^2 - \sum_{j=1, j \neq i}^N K_{eij} \eta_{eij}^2 - K_{2i} z_{2i}^T z_{2i} \right. \\ &\quad \left. - \frac{\sigma_{\delta i}}{2n_{\delta i}} \tilde{\delta}_i^2 + c_{1i} \right), \end{aligned} \quad (39)$$

where  $\tilde{\delta}_i = \hat{\delta}_i - \bar{\delta}_i$  ( $i = 1, \dots, N$ ), and  $c_{1i}$  is a constant defined as  $c_{1i} \triangleq \varepsilon_i (1 + \bar{\delta}_i) + \frac{\sigma_{\delta i}}{2n_{\delta i}} \bar{\delta}_i^2$ .

Denote  $V_{\text{UAV1}} = V_{\text{Load1}} + V_{\text{Load2}} + V_{\text{pos}}$ , after some algebraic manipulation, we can get

$$\dot{V}_{\text{UAV1}} < -\kappa_1 V_{\text{UAV1}} + \varrho_1, \quad (40)$$

where  $\kappa_1 \triangleq \min_{i,j} (2K_{1L}, 2K_{2L}, 2K_{3L}, 2K_{4L}, 2K_{ei}, 2K_{eij}, 2K_{2i}, \sigma_{\delta 1L}, \sigma_{\zeta L}, \sigma_{\rho JL}, \sigma_{\delta 2L}, \sigma_{\delta i})$ ,  $\varrho_1 \triangleq c_{1L} + c_{2L} + \sum_{i=1}^N c_{1i}$ .

**Theorem 4.1:** With the UAV thrust law designed as (37), and adaptive laws (22), (28), (29), (30), and (38), the quadrotor system and the payload system described in (1) and (2) under Assumptions 2.1–2.5 have the following properties:

- The constraint requirements (10), (11), and (12) will not be violated during operation.
- The transformed distance tracking errors  $\eta_{eL}$ ,  $\eta_{ei}$ ,  $\eta_{eij}$ , ( $i = 1, \dots, N, j \neq i$ ) and angle tracking error of

payload  $z_{3L}$  will converge into the sets

$$\left\{ x = \eta_{eL}, \eta_{ei}, \eta_{eij} \mid |x| < \varepsilon_\eta, \varepsilon_\eta = \sqrt{\frac{2\rho_1}{\kappa_1}} \right\}, \quad (41)$$

$$\left\{ z_{3L} \mid \|z_{3L}\| < \varepsilon_\eta, \varepsilon_\eta = \sqrt{\frac{2\rho_1}{\kappa_1}} \right\}, \quad (42)$$

and as a result, the actual distance tracking errors  $d_{eL}$ ,  $d_{ei}$ , and  $d_{eij}$  will converge to the sets

$$\left\{ x = d_{eL}, d_{ei} \mid x < \varepsilon_{\chi_{H,i}} \right\}, \quad (43)$$

$$\left\{ d_{eij} \mid -\varepsilon_{\iota_{L,i}} < d_{eij} < \varepsilon_{\iota_{H,i}} \right\}, \quad (44)$$

where

$$\varepsilon_{\chi_{H,i}} = \frac{\varepsilon_\eta \Omega_{dHi}}{\Omega_{dHi} + \varepsilon_\eta}, \quad (45)$$

and we have  $\varepsilon_{\iota_{H,i}}$  expressed in (46), with  $\varepsilon_{\iota_{W,i}}$  expressed in (47) (see the next page), where  $\Omega_H = \Omega_{Hi}$  and  $\Omega_W = \Omega_{Wij}$  for  $i, j = 1, \dots, N, j \neq i$ .

*Proof:* Omitted due to space limit. ■

## V. SIMULATION STUDIES

In this section, a simulation example is carried out with a team of  $N = 4$  quadrotors and a cuboid payload. In this simulation, the model parameters of the quadrotors are  $m_i = 4\text{kg}$ ,  $g = 9.81\text{m/s}^2$ , and  $J_i = \text{diag}[0.109, 0.103, 0.0625]\text{kg} \cdot \text{m}^2$ ,  $i = 1, 2, 3, 4$ . The mass of the payload is  $m_L = 2\text{kg}$  and its length, width, and height are 1.0m, 0.8m, and 0.2m, respectively. The length of cables is  $l_i = 3.5\text{m}$ ,  $i = 1, 2, 3, 4$ , and they are attached to the points of the payload  $\rho_1 = [0.5, 0.4, 0]^T$ ,  $\rho_2 = [0.5, -0.4, 0]^T$ ,  $\rho_3 = [-0.5, -0.4, 0]^T$ , and  $\rho_4 = [-0.5, 0.4, 0]^T$ . The desired trajectory of the payload is selected as  $p_{dL} = [0.25 \cos(0.25t) - 0.65, 0.25 \sin(0.25t) + 0.15, -0.3t + 1.5]^T$ . The desired attitude of the payload is  $\Theta_{dL} = [0, 0, 0]^T$ . The desired unit direction vector from the mass center of the  $i$ th UAV towards the  $i$ th link attachment point is selected as  $e_{di} = [0, 0, 1]^T$ ,  $i = 1, 2, 3, 4$ . Thus, the desired trajectory of the  $i$ th UAV can be written as  $p_{di} = p_{dL} + R(\Theta_{dL})\rho_i - l_i e_{di}$ . The constraint functions are selected as  $\Omega_{dHi} = (2.5 - 0.07)e^{-0.3t} + 0.07$ ,  $\Omega_{dHL} = (2.5 - 0.07)e^{-0.3t} + 0.07$ ,  $\Omega_{Hij} = (0.8 - 0.05)e^{-0.1t} + 0.05$ , and  $\Omega_{Wij} = (0.8 - 0.05)e^{-0.1t} + 0.05$ ,  $i, j = 1, 2, 3, 4, i \neq j$ . The initial position and attitude of the payload are  $p_L(0) = [0, 0, 0]^T$  and  $\Theta_L(0) = [0.1, 0.1, 0.1]^T$ , respectively. Thus, the initial position of the  $i$ th UAV can be expressed as  $p_i(0) = p_L(0) + R(\Theta_L(0))\rho_i - l_i e_i(0)$ , where  $e_i(0) = [0, 0, 1]^T$ , and the initial attitude of the  $i$ th UAV is  $\Theta_i(0) = [0, 0, 0.1]^T$ ,  $i = 1, 2, 3, 4$ . The initial velocities of every UAV and payload are zero.

The LOS distance tracking errors  $d_{ei}$  and  $d_{eL}$  under the proposed controller are shown in Fig. 1 with  $\Omega_{dHi}$  and  $\Omega_{dHL}$ ,  $i = 1, 2, 3, 4$ . From this figure, we see that  $d_{ei}$  and  $d_{eL}$  can converge to a small neighborhood of the origin without violation of the performance constraints  $\Omega_{dHi}$  and  $\Omega_{dHL}$ , respectively. Fig. 2 gives the profile of the inter-quadrotor distance tracking errors  $d_{eij}$  under the proposed controller,  $i, j = 1, 2, 3, 4, j \neq i$ . It is obvious that the safety constraints are always satisfied during the cooperative transportation operation since  $d_{eij}$  always stays between the constraint functions,  $-\Omega_{Wij}$  and  $\Omega_{Hij}$ . Based on the above discussion, we can conclude that the simulation results confirm the theoretic analysis shown in Theorem 4.1.

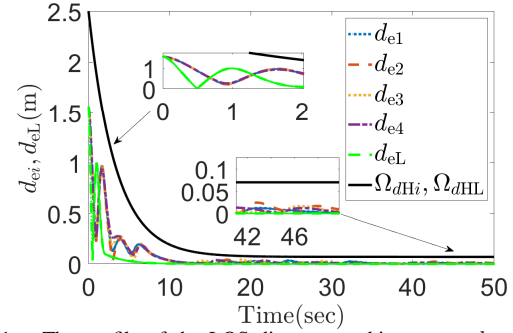


Fig. 1. The profile of the LOS distance tracking errors  $d_{ei}$  with  $\Omega_{dHi}$ ,  $i = 1, 2, 3, 4$ , and  $d_{eL}$  with  $\Omega_{dHL}$ .

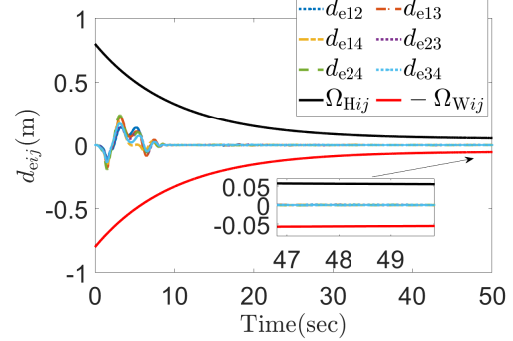


Fig. 2. The profile of the relative inter-quadrotor distance tracking errors  $d_{eij}$  with  $\Omega_{Hij}$  and  $-\Omega_{Wij}$ ,  $i, j = 1, 2, 3, 4, i \neq j$ .

## VI. CONCLUSION

In this work, we propose a new adaptive constrained formation architecture for cooperative load transportation by a group of UAVs. Multiple performance and safety constraint requirements during the operation are dealt with by the universal barrier function structures. Exponential convergence of the distance and attitude tracking errors can be guaranteed. Future research includes experiment verification of the framework, and resiliency study in face of random sensor-actuator attacks and system faults during the transportation.

## APPENDIX

### A. Step 4 of Backstepping Design

Taking derivative of  $V_{4L}$  leads to

$$\begin{aligned} \dot{V}_{4L} = & z_{4L}^T \left( \Psi_L^T J_L^T \sum_{i=1}^N \mathbb{S}(r_i) (T_{di} e_{di} - T_i e_i) \right. \\ & \left. + \Psi_L^T J_L^T \sum_{i=1}^N \mathbb{S}(r_i) (-T_{di} e_{di}) - M_L \dot{\alpha}_{\Theta_L} - C_L \alpha_{\Theta_L} \right), \end{aligned} \quad (48)$$

where  $\frac{1}{2} z_{4L}^T (\dot{M}_L - 2C_L) z_{4L} = 0$ . Under Assumption 2.2 and Lemma 2.1, from (48) we have

$$\begin{aligned} & z_{4L}^T \Psi_L^T J_L^T \sum_{i=1}^N \mathbb{S}(r_i) (T_{di} e_{di} - T_i e_i) \\ & \leq \|z_{4L}\| \bar{\delta}_{2L} < \varepsilon_L \bar{\delta}_{2L} + \bar{\delta}_{2L} \frac{\|z_{4L}\|^2}{\sqrt{\|z_{4L}\|^2 + \varepsilon_L^2}}, \end{aligned} \quad (49)$$

$$\varepsilon_{\iota_{H,i}} = \frac{-(\Omega_H \Omega_W - \varepsilon_\eta(\Omega_H - \Omega_W)) + \sqrt{\Omega_H^2 \Omega_W^2 + \varepsilon_\eta^2(\Omega_H + \Omega_W)^2 - 2\varepsilon_\eta \Omega_H \Omega_W(\Omega_H - \Omega_W)}}{2\varepsilon_\eta} \quad (46)$$

$$\varepsilon_{\iota_{W,i}} = \frac{(\Omega_H \Omega_W + \varepsilon_\eta(\Omega_H - \Omega_W)) - \sqrt{\Omega_H^2 \Omega_W^2 + \varepsilon_\eta^2(\Omega_H + \Omega_W)^2 + 2\varepsilon_\eta \Omega_H \Omega_W(\Omega_H - \Omega_W)}}{2\varepsilon_\eta} \quad (47)$$

where  $\bar{\delta}_{2L} > 0$  is an *unknown* constant. Furthermore, using the notation definitions in (5), we can obtain the following

$$\begin{aligned} & z_{4L}^T (-M_L \dot{\alpha}_{\Theta L} - C_L \alpha_{\Theta L}) \\ & \leq \|z_{4L}\| \|\Psi_L\| \|J_L\|^2 \left( \left\| \Psi_L \dot{\alpha}_{\Theta L} + \dot{\Psi}_L \alpha_{\Theta L} \right\| \right. \\ & \quad \left. + \left\| \mathbb{S}(\Psi_L \dot{\Theta}_L) \right\| \|\Psi_L \alpha_{\Theta L}\| \right) \\ & < \varepsilon_L \bar{\zeta}_L + \bar{\zeta}_L \frac{\|z_{4L}\|^2 \Xi_L^2}{\sqrt{\|z_{4L}\|^2 \Xi_L^2 + \varepsilon_L^2}}, \end{aligned} \quad (50)$$

where  $\bar{\zeta}_L \triangleq \|J_L\|^2$  is *unknown*, and  $\Xi_L \triangleq \|\Psi_L\| \left( \left\| \Psi_L \dot{\alpha}_{\Theta L} + \dot{\Psi}_L \alpha_{\Theta L} \right\| + \left\| \mathbb{S}(\Psi_L \dot{\Theta}_L) \right\| \|\Psi_L \alpha_{\Theta L}\| \right)$  is known.

### B. Step 6 of Backstepping Design

The derivative of  $V_1$  with respect to time leads to  $\dot{V}_1 = \sum_{i=1}^N (\eta_{ei} \dot{\eta}_{ei} + \sum_{j=1, j \neq i}^N \eta_{ej} \dot{\eta}_{ej})$ . First we examine the dynamics for  $\eta_{ei}$  ( $i = 1, \dots, N$ ). From (14), we have

$$\begin{aligned} \dot{\eta}_{ei} &= \frac{\partial \eta_{ei}}{\partial d_{ei}} \frac{1}{d_{ei}} (x_i - x_{di}) \dot{x}_i + \frac{\partial \eta_{ei}}{\partial d_{ei}} \frac{1}{d_{ei}} (y_i - y_{di}) \dot{y}_i \\ &+ \frac{\partial \eta_{ei}}{\partial d_{ei}} \frac{1}{d_{ei}} (z_i - z_{di}) \dot{z}_i - \xi_i, \end{aligned} \quad (51)$$

where  $\xi_i \triangleq \frac{\partial \eta_{ei}}{\partial d_{ei}} \frac{1}{d_{ei}} (x_i - x_{di}) \dot{x}_{di} + \frac{\partial \eta_{ei}}{\partial d_{ei}} \frac{1}{d_{ei}} (y_i - y_{di}) \dot{y}_{di} + \frac{\partial \eta_{ei}}{\partial d_{ei}} \frac{1}{d_{ei}} (z_i - z_{di}) \dot{z}_{di} - \frac{\partial \eta_{ei}}{\partial \Omega_{dHi}} \dot{\Omega}_{dHi}$ .

For  $\dot{V}_{ej}$  ( $i, j = 1, \dots, N, j \neq i$ ) we have

$$\begin{aligned} \dot{V}_{ej} &= \eta_{ej} \frac{\partial \eta_{ej}}{\partial d_{ej}} \frac{1}{d_{ij}} (x_i - x_j) \dot{x}_i + \eta_{ej} \frac{\partial \eta_{ej}}{\partial d_{ej}} \frac{1}{d_{ij}} (y_i - y_j) \dot{y}_i \\ &+ \eta_{ej} \frac{\partial \eta_{ej}}{\partial d_{ej}} \frac{1}{d_{ij}} (z_i - z_j) \dot{z}_i - \eta_{ej} \frac{\partial \eta_{ej}}{\partial d_{ej}} \frac{1}{d_{ij}} (x_i - x_j) \dot{x}_j \\ &- \eta_{ej} \frac{\partial \eta_{ej}}{\partial d_{ej}} \frac{1}{d_{ij}} (y_i - y_j) \dot{y}_j - \eta_{ej} \frac{\partial \eta_{ej}}{\partial d_{ej}} \frac{1}{d_{ij}} (z_i - z_j) \dot{z}_j \\ &- \eta_{ej} \xi_{ej}, \end{aligned} \quad (52)$$

where  $\xi_{ej} \triangleq \frac{\partial \eta_{ej}}{\partial d_{ej}} \frac{1}{d_{ij}} (\dot{x}_{di} - \dot{x}_{dj}) (x_{di} - x_{dj}) + \frac{\partial \eta_{ej}}{\partial d_{ej}} \frac{1}{d_{ij}} (\dot{y}_{di} - \dot{y}_{dj}) (y_{di} - y_{dj}) + \frac{\partial \eta_{ej}}{\partial d_{ej}} \frac{1}{d_{ij}} (\dot{z}_{di} - \dot{z}_{dj}) (z_{di} - z_{dj}) - \left( \frac{\partial \eta_{ej}}{\partial \Omega_{Hi}} \dot{\Omega}_{Hi} + \frac{\partial \eta_{ej}}{\partial \Omega_{Wij}} \dot{\Omega}_{Wij} \right)$ .

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